complicated scientific theories that are inarguably correct. “Here’s a ‘multiverse’ that basically everyone believes in,” says David Wallace, a philosopher and advocate of the many-worlds interpretation. “Think about the planets of the stars of distant galaxies. Pretty much everyone thinks that there are indeed planets around the stars in distant galaxies, and that there are rocks on the surface of those planets.… That’s not an infinite multiverse, but [ten thousand billion billion] solar systems is quite a lot to be getting on with. And the reason you take that seriously is not really because we can observe it.… It’s more that it’s a completely unavoidable consequence of a theory that we think is just rock-solid.”

Physicists attacking many-worlds (or inflation, or string theory) usually have a more serious objection to the idea of a multiverse: they denounce it as a prime example of “unfalsifiability.” This unwieldy word, a ghost of philosophy past, comes from the work of Karl Popper. Popper was a celebrated mid-twentieth-century philosopher of science who spent most of his career at the London School of Economics. Popper had once held truck with the logical positivism of his native Vienna but ultimately took an iconoclastic stance of his own. Rather than championing a verification theory of meaning, as the Vienna Circle had done, Popper promoted a scientific worldview based on *falsification.* Theories that could be proven false, Popper declared, were potentially scientific theories—and theories that could not be proven false were not scientific at all.

Popper’s views became unusually popular among practicing scientists, and, by the end of the twentieth century, many physicists believed that falsifiability was a vital acid test that any potential theory must pass. Viewed through this lens, any multiverse theory appears quite suspect. If the other universes are not accessible and can never directly influence our own universe, then what possible experimental data could falsify the theory that we live in a multiverse? And if no data could ever show the theory wrong, then how can it be an acceptable scientific theory? “As the philosopher of science Karl Popper argued: a theory must be falsifiable to be scientific,” wrote prominent cosmologists George Ellis and Joe Silk in a *Nature* editorial in 2014. “These unprovable hypotheses [many-worlds, string theory, and the inflationary multiverse] are quite different from those that relate directly to the real world and that are testable through observations—such as the standard model of particle physics and the